

YEN-LIN (ALLEN) LU

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EDUCATION

New York University Tandon School of Engineering	May 2026
<i>Bachelor of Science in Mathematics & Computer Science; Dean's List for all academic years</i>	GPA: 3.73/4.00
University of California, Irvine	Sep 2022 – Jun 2023
<i>Bachelor of Science in Mathematics; Dean's Honor List for all quarters</i>	GPA: 3.99/4.00

TECHNICAL SKILLS

Languages & Tools: Python, C++, C, R, SQL, JavaScript, Java, HTML/CSS, Node.js, LaTeX, Git/GitHub, MATLAB, GSuite, MongoDB
Frameworks & Libraries: PyTorch, scikit-learn, XGBoost, imbalanced-learn, Pandas, NumPy, Matplotlib, Express.js, Bootstrap, React, Jest, Mongoose

EXPERIENCE

Research Assistant	Oct 2025 – Present
<i>NYU Cognition and Computation Lab</i>	<i>New York, NY</i>
- Develop dual LLM-Bayesian learning system translating natural language hints into symbolic meaning functions using PyTorch, Transformers, and LLMs for explainable cognitive modeling - Implement in-context learning pipelines and benchmark experiments evaluating LLM-generated vs. hand-coded meaning functions for probabilistic inference	
Student Product Support Analyst	Oct 2024 – Present
<i>New York University</i>	<i>New York, NY</i>
- Test and troubleshoot NYU Brightspace updates pre-deployment, ensuring smooth transitions for 80,000+ users - Develop training materials and documentation, manage user support, and deliver live Brightspace training sessions	
Software Development Engineer Intern	Jun 2024 – Aug 2024
<i>Athleo.ai</i>	<i>Boston, MA (Remote)</i>
- Engineered a responsive website using HTML, CSS, and JavaScript, attracting 1,500+ daily views - Executed QA testing with software team, resolving 20+ client-reported issues and implementing user-driven improvements that increased reliability and user satisfaction - Produced and optimized technical videos, achieving 30,000+ views and streamlining product onboarding	

PROJECTS

IAL Trading Platform <i>JavaScript, Node.js, Express.js, MongoDB, Bootstrap, Jest</i>	Sep 2025 – Present
- Built full-stack stock trading platform with fractional share trading, portfolio analytics, and profit/loss tracking - Architected Express.js RESTful API with MongoDB backend and implemented bcrypt authentication with responsive Bootstrap UI - Wrote 33+ Jest unit tests achieving 100% coverage for trading validation logic and financial calculations	
NBA Championship Prediction System <i>Python, PyTorch, Sklearn, Pandas, NumPy</i>	May 2025 – Aug 2025
- Engineered an end-to-end sports analytics pipeline, integrating data preprocessing, feature engineering, probability calibration, and AWS S3 model hosting to predict NBA championship probability with 91% accuracy - Implemented and benchmarked multiple ML models (PyTorch neural network, logistic regression, random forest, XGBoost) with calibration and soft-vote ensembling, improving F1 score by 15% - Built an interactive Streamlit web app enabling users to construct custom rosters under salary-cap constraints and view real-time championship predictions	
Loan Default Predictor <i>Python, Sklearn, NumPy, Matplotlib</i>	Jan 2025 – May 2025
- Optimized models (logistic regression, SVM, neural network) to predict loan defaults with 92.1% accuracy - Preprocessed 32K+ data entries using imputation, one-hot encoding, scaling, and PCA to reduce bias by 20% - Improved F1 score by 12% through targeted feature engineering, hyperparameter tuning, and model optimization	
Airline Ticket Reservation System <i>Python, Flask, SQL, HTML, CSS</i>	Sep 2024 – Jan 2025
- Designed an online flight booking system with distinct functionalities and UI for customers and airline staff accounts - Architected a Flask-based API with MySQL backend, handling concurrent requests and secure database transactions	

LEADERSHIP & INVOLVEMENT

Applied Math Club – Co-Founder, Secretary, Treasurer | **NYU Theta Tau Professional Engineering Fraternity** – Webmaster, Treasurer | **NYU VIP RoboMaster Team** – Software Team Member